SYS 3060: A01

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Question 1

A student will run experiments in a lab session. With a probability of 0.5, the student will have enough time to run only one experiment, and with a probability of 0.5, the student will have enough time to run two experiments. Each experiment will fail with probability 0.3, independently of the other experiments.

(a) Find the probability that the student succeeds in all experiments during a lab session.

If they run one experiment, that one must succeed. If they run two experiments, *both* must succeed.

Case 1: Each experiment fails with probability 0.3, so succeeds with 0.7. This happens with probability 0.5 meaning this case contributes 0.5×0.7.

Case 2: The two experiments succeed *independently* with probability 0.7. This situation happens with probability 0.5 contributing $0.5 * 0.7^2$.

Thus, the probability that the student succeeds in all experiments is 0.595.

□

(b) Find the probability that exactly one experiment fails.

```
ans =  0.5*0.3 + 0.5*(0.3*0.7 + 0.7*0.3)
```

Using a case by case deconstruction similar to (a), we get 0.36.

□

(c) Find the probability that the student performs two experiments and both succeed.

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0.5 * (0.7 * 0.7)
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0.24499999999999997
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□

(d) Given that the first experiment succeeds, find the probability that the second experiment fails.

Due to the independence,

$$P(\text{second fails} \mid \text{first succeeds}) = P(\text{second fails}) = 0.3$$

□

(e) Find the expected number of successful experiments in a lab session.

In Case 1, the expectation of X given that there is only one experiment is 0.7 as given. For cae 2, it would be $0.7 + 0.7 = 1.4$. Thus we get

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ans = 0.5*0.7 + 0.5*1.4
```

which yields 1.0499999999999998. Which makes sense because on average the student runs 1.5 experiments and $0.5 \times 0.7 = 1.05$.

□

(f) Find the variance of the number of experiments the student runs in a lab session.

Define N to be the *number of experiments run in a lab session*, where $N = 1$ with half probabiliy and $N = 2$ with half probability.

We know that

$$E(N) = 1(0.5) + 2(0.5) = 0.5 + 1 = 1.5,$$

which is a fact we used in the previous part as well.

Similarly we can also conclude that

$$E(N^2) = 1(0.5) + 4(0.5) = 2.5.$$

Thus,

$$\text{Var}(N) = E(N^2) - (E(N))^2$$

which is 0.25.

□

Question 2

A lab sensor detects particles. The time X (in minutes) until the nexxt detection is exponential with rate λ.

(a) Find the time τ such that there is 75% chance that the sensor detects a particle by time τ.

We are told that $X \sim$ Exponential for which we know the CDF to be

$$P(X \le t) = 1 - e^{-\lambda t}, \quad t \ge 0.$$

We just want to solve for

$$1 - e^{\lambda\tau} = 0.75$$

Which yields the following answer:

$$\begin{aligned} e^{-\lambda t} &= 0.25 \\ -\lambda\tau &= \ln(0.25) \\ \tau &= -\ln(0.25)/\lambda \\ \tau &= \ln(4)/\lambda \end{aligned}$$

□

(b) Find the probability that the waiting time until the next particle detection exceeds 5 minutes.

We want $P(X > 5) = e^{-5\lambda}$. That's the answer.

□

(c) Find the probability that a particle is detected between the next 5 to 8 minutes.

$P(5 < X < 8) = (1 - e^{-8\lambda}) - (1 - e)^{-5\lambda} = e^{-5\lambda} - e^{-8\lambda}$.

□

(d) Given that the detection occured in the first ten minutes, find the probability that another detection occurs within the next 2 minutes.

Given that the exponential distribution is memoryless, the future waiting time does not depend on when the previous detection occurred. So the answer is just

$$1 - e^{-2\lambda}.$$

□

(e) Find the expected waiting time until the next particle is detected and its variance, consider $\lambda = 0.1$ detections/minute.

Well the expectation and variance are described by

$$\mathbb{E}(X) = 1/\lambda, \quad \mathbb{V}(X) = 1/\lambda^2.$$

This gives us answers, 10 minutes and 100 minutes^2, respectively.

□

Question 3

Students arrive at a shuttle top according to a Poisson process with rate

$$\lambda = \text{ 6 students per hour.}$$

Each arriving student is an undergraduate with probability 0.7 and a graduate with probability 0.3, independently of everything else. Let $N(t)$be the total number of arrivals in the firt t hours. Let $N_U(t)$ and $N_G(t)$ be the numbers of undergraduates and graduates arriving in the first t hours.

(a) Compute $P(N(0.5) = 2)$.

For $N(t) \sim \text{Poisson}(\lambda t)$,

$$\lambda t = 6 \times 0.5 = 3 \implies N(0.5) \sim \text{Poisson}(3).$$

We know that

$$P(N = k) = \frac{\mu^k e^{-\mu}}{k!}$$

which when $k = 2$ and $\mu = 3$ are subbed in, we get

$$P(N(0.5) = 2) = 4.5e^{-3}.$$

□

(b) Compute $P(N_U(1) = 3$ and $N_G(1) = 1)$.

Because the two counts are independent of each other, we can say:

$$P(N_U(1) = 3 \cap N_G(1) = 1) = P(N_U(1) = 3) \cdot P(N_G(1) = 1).$$

Just compute both and multiply.

```
import math

lambda_u = 4.2   # UG/hour
lambda_g = 1.8   # G/hour

# Probabilities
p_u = (lambda_u**3 * math.exp(-lambda_u)) / math.factorial(3)
p_g = (lambda_g**1 * math.exp(-lambda_g)) / math.factorial(1)

p = p_u * p_g
```

By hand I arrived to $(4.2^3)(1.8)(e^{-6})/3!$ which does match our computed answer of 0.05509373737945715.

□

(c) Given that $N(1) = 4$, compute the probability that exactly 3 of them are undergradutes.

So here we are just looking at total count, the Poisson distribution doesn't matter because we are classifying four independent arrivals. Given $N(1) = 4$ we know

$$N_U(1) \mid N(1) = 4 \sim \text{Bin}(4, 0.7).$$

We want to know $P(N_U(1) = 3 \mid N(1) = 4)$ which is simply

$$\binom{4}{3}(0.7)^3(0.3)^1$$

which is 0.4115999999999999.

$\square$

Question 4

A device experiences failures over time. The number of failures observed up to time t is modeled by a non-homogeneous Poisson process (NHPP) $\{N(t), t \geq 0\}$ with time-varying rate

$$\lambda(t) = 3e^{-t}, \quad t \geq 0,$$

where t is measured by days and $\lambda(t)$ has units of failures per day. Let $N(t)$ denote the total number of failures that occur in the same time interval $[0, t]$.

Let T_1 denote the time of the first failure:

$$T_1 = \inf\{t \geq 0 \mid N(t) \geq 1\}.$$

Recall that the CDF of T_1 is defined as

$$F_{T_1}(t) = P(T_1 \geq t), \quad t \geq 0,$$

which represents the probability that the first failure occurs by time t. *Hint: You may ue the NHPP mean value function*

$$\Lambda(t) = \int_0^t \lambda(s) ds.$$

(a) Compute the probability of no failures occur during the interval $[0, 2]$, i.e., compute $P(N(2) = 0)$.

$$\Lambda(t) = \int_0^t \lambda(s)ds$$

So $P(N(t) = 0) = \exp(-\Lambda(t))$ where $\lambda(t) = 3e^{-t}$, so

$$\Lambda(2) = \int_0^2 3e^{-s}ds = 3(1 - e^{-2}).$$

Therefore,

$$P(N(2) = 0) = \exp(-3(1 - e^{-2})).$$

In decimal form this is 0.07472099638076134.

□

(b) Derive an explicit expression for the CDF of T_1, i.e., find $F_{T_1}(t) = P(T_1 \geq t)$ for $t \geq 0$.

The event *the first failure happens after time t* is the same as *no failures occur by time t* which is denoted: $\{T_1 > t\} \iff \{N(t) = 0\}$.

Similar to above we derive that

$$P(T_1 > t) = \exp(-3(1 - e^{-t})).$$

And, we know the actual form of a CDF is $P(T_1 \geq t)$ in the situation, yielding

$$P(T_1 \geq t) = 1 - \exp(-3(1 - e^{-t})).$$

Small change to do it with the other sign.

□

(c) Compute $P(T_1 > 1)$, the probability that the first failure occurs after day 1.

We already have $P(T_1 > t) = \exp(-3(1 - e^{-1}))$, so with $t = 1$ we get 0.15011378939830683.

□